Every fix the audit supports, for an airtight v24

This list comes from the full text of all ten reports: the five primary-source audits (A to E), the three conceptual exhibit checks and the two Exhibit 4 checks. It is a list only, and nothing has been changed. The yellow cells are proposed wording, built only from passages the reviewers quoted from the sources. Tier 1 is the thesis. Tier 2 is plain factual errors. Tier 3 is overstatements that a knowledgeable reader would catch. Tier 4 is precision and nits. Section 5 lists evidence the report does not use but could.

Tier 1. The thesis and its headline claims

#	Where	Problem (source)	Fix
1	Cover, p2, p12	"Clients' AI tools could act on Aladdin under the same controls BlackRock uses for its own business." This merges the API-level controls, which any caller faces, with Copilot's guardrails, which sit inside Copilot (Audits A and B, conceptual check 13–17).	Evolve the thesis. "Clients' AI tools could call the same governed tools Copilot calls, under the same permissions and API controls." Keep the MCP route as a labelled inference.
2	p2	"BlackRock's advantage is that it already runs its own business this way." No source shows Copilot acting. Valdez: "hopefully next year, we are here doing autonomous agents."	"BlackRock already runs AI on these APIs. Copilot answers questions for staff and clients by calling about two thousand tools and agents, most mapped to production Aladdin APIs. Public sources show it answering, not acting."
3	p2 "Aladdin will not be alone"; p12 "unlikely to stay unique"	This has the competitive position the wrong way round. Arcesium's MCP "deploys connectors, builds pipelines, and creates quality rules". Clearwater runs 800+ agents "ready to act". SimCorp has a portfolio management agent live today. The executive summary also hardens Part 3's "unlikely" (Audit E).	"Rivals have begun letting AI agents act: Arcesium through an MCP server, Clearwater and SimCorp inside their platforms. BlackRock already runs AI over the production APIs and controls it sells."
4	p2 "Many data providers now offer AI tools a way to read their data. Aladdin's APIs can also write"	The contrast fails for Arcesium, whose MCP server writes (Audit E).	"Many data providers now let outside AI tools read their data, and some let them act. Aladdin's APIs can write: they post orders and assign compliance rules."
5	p2 "Access is granted per API, to read or to write"; p5 "per API and per action"	17 of 54 specs use one scope for the whole API, and 3 mix the two styles. That leaves 96 of 301 operations with no read/write split. No scope is per operation. Client-key and basic-auth calls carry no scopes (Audits A and C; I re-ran the count).	p2: "Each API has its own OAuth scopes, and most split read from write." p5: delete "and per action". Add: "Aladdin user permissions add limits by function, action and portfolio" (the PermissionAPI and per-operation notes, such as permType='brkrSettlm').
6	p2, p12 compliance sentences	Correct but incomplete. It leaves out `complianceTimeout`: "orders will be posted with the 'Compl Pending' status set regardless of any violations". Posting needs the same `Order:write` scope with or without the check. TradeAPI posting has no compliance option. Violations come back with the created order, and the spec does not show them blocking it (Audits A and C).	"Posting an order runs compliance by default. The caller can turn it off with the same scope it uses to post, and a timeout can post the order before the check finishes. Whether Aladdin user permissions restrict this is not public. Trades posted directly have no compliance setting."

7	p2 "will grow 17x"; p8 attribution; p8 heading; p9 box; p12	Slide 23 is titled "Continuously drive marginal costs down". The figure is a "2030 Estimate" for "BlackRock + Aladdin", with "no guarantee", no base year and no caller named. "Studio" is not in the deck. The p9 box drops the "I attribute" hedge: "two growth bets both run through Studio" (Audit D, conceptual check 7–12; I checked slide 23).	p2: "BlackRock estimated that API calls across BlackRock and Aladdin would increase 17x by 2030." p8: replace "I attribute those API calls to Studio…" with "The slide does not say who makes these calls. They include BlackRock's own applications and Copilot." p8 heading and p9 box: drop "runs through Studio". Use slide 33, "Growing Aladdin as a Platform", slide 35's "Aladdin as a platform" growth driver, and the 10-K's "empowering clients with data and APIs".
8	p9, p12, Exhibit 14 "How they would connect is not published"	This blurs three things: (a) technical access, (b) documented AI support and (c) a published connector. Licensed clients can already call the APIs with a Studio OAuth app, including the no-user `client_credentials` flow. The specs mention "bot accounts". Only (b) and (c) are missing (Audit B).	"Any client program, an AI agent included, can already call the APIs with Studio-registered credentials. BlackRock has published no support, guidance or connector for AI agents."
9	Exhibit 14 hatched box and note; Exhibit 12 row; p9	The job posting is called the "nearest public sign". It is for BlackRock's internal Enterprise Data Platform, with MCP listed "to automate repetitive onboarding tasks". A much nearer sign is in source 71: Garson says "100%" and "we are hearing that from our clients" when asked about clients wanting Aladdin plug-ins in their own copilots (Audits B and E; I checked the transcript).	Use the source 71 exchange as the public sign of demand. Describe the posting accurately, or drop it.
10	Exhibit 15 row 2	"under the same permissions, compliance checks and guardrails Copilot uses". Copilot's guardrails do not apply to a client's AI tool.	Make the row the evolved proposal: "A governed route for clients' AI tools: the same registered tools and permissions Copilot uses, with approval before any order posts."
11	Exhibit 14 (whole)	The left column is headed "BlackRock's own use", but clients use the apps and Copilot. The legend covers boxes, not arrows, so the inferred Copilot-to-public-API link is drawn solid. The API controls panel sits under Copilot, with no evidence Copilot goes through OAuth scopes. The Copilot guardrails sit under the client column. The registry's per-tool gating, which is the control that matters most, is missing (Audits A and B, conceptual check 13–17).	Redraw for the evolved thesis. Top: "Built by BlackRock (used by BlackRock and clients)" beside "Built by the client". Middle: shared server-side layer (tools and registry permissions, API scopes, Aladdin user permissions, compliance on orders). Bottom: what stays inside Copilot (input and output guardrails, planning, AI-graded tests). The client AI route is dashed and marked as proposed. The legend covers arrows.
12	Exhibit 3 layering (p4)	Stacking Studio under BlackRock's applications says the apps run on Studio, and that clients build on the apps. Khara: Studio is "over and above what they get through our front end applications". Exhibit 14 draws the two side by side. Four of six panellists flagged this (conceptual check 1–6).	Keep your layered design, but put "BlackRock's applications and Copilot" and "Aladdin Studio" side by side in one row over the shared services. Point "Who builds" at Studio only.

Tier 2. Plain factual errors

#	Where	Problem	Fix
13	Exhibit 2, Order "Where" (p3)	"Explore what-if". Explore's what-if compares hypothetical portfolios for risk. The OrderAPI spec puts orders in the "Order Monitor of Aladdin Dashboard".	"Aladdin Dashboard (Order Monitor)". Drop source 27 from this cell.

14	p9 and Exhibit 12 note	"Its only public mention of MCP is a job posting." BlackRock's webinar moderator says "You can have your MCP, you can have your different APIs" (source 10).	"BlackRock has published no MCP server or connector for Aladdin."
15	p14; Exhibit 17 Arcesium "first call" = 0	Source 96: "request a 30-day trial … There are no sales calls required to get started." Pricing is $2,000–$4,000 a month.	Remove Arcesium from the "needs a contract" list. Score its first call 1 (and its trial 1 or 2). Re-total.
16	Exhibit 12 title (p10)	"Ten of the 11 … announced AI features or AI access". Aladdin meets that test with Copilot, eFront Copilot and Auto Commentary.	For example: "Most of the 11 offer outside AI tools a connector or MCP server. Aladdin, SimCorp and Clearwater offer AI inside their platforms." Verify the count before using "most".
17	Exhibit 12 Aladdin row	Shows only the job posting, while rivals' in-platform agents count.	"Copilot and eFront Copilot for clients, inside the platform. No published connector or MCP server for outside AI tools."
18	Exhibit 17 title (p14)	"Data platforms open to any developer", yet FactSet and MSCI score 0 for a first call.	"Snowflake and Databricks are self-serve. Most others, Studio included, open fully only to clients."
19	Exhibit 2, Valuation detail	"Each valuation carries an exception list". It carries a count, and a separate call lists the exceptions.	"Each valuation records how many exceptions it raised. A separate call lists them."
20	Exhibit 4 SDK tile	"typed clients for the APIs and Data Cloud". The Data Cloud client is "a wrapper around snowflake-connector-python" that takes SQL. Rate limiting applies to the APIs only.	"Open-source Python library with typed clients for the APIs and a Snowflake connection for Data Cloud, with login, retries and API rate limits built in."
21	Exhibit 6, Miller cell	Data Cloud "replaced" flat files and FTP. He said it "makes it much simpler". This also clashes with ADX in Exhibit 4.	"Data Cloud is simpler than the flat files and FTP used before."

Tier 3. Overstatements a knowledgeable reader would catch

#	Where	Problem	Fix
22	Exhibit 2 title	"cover a trade from order to NAV". Writes stop at orders and trades. Nothing covers execution, confirmation, settlement or positions. Lots and NAV are read-only.	"The public APIs post orders and trades and read what follows, except positions."
23	Exhibit 2, Compliance "Who: No one"	Violations have owners, dispositions and an optional four-eyes flag. The check runs inside the OrderAPI post, not in ComplianceRuleAPI or ViolationAPI.	Who: "The engine runs it; a named team resolves violations." API: "OrderAPI (runCompliance); ViolationAPI to resolve".
24	Exhibit 2, Position	"Where: Data Cloud" reads as the book of record. "Positions appear only as fields" undersells two APIs: CollateralStatementAPI has a full Position object, and CashLadderAPI returns cash balances.	Where: "Read from Data Cloud (a copy of the IBOR)". Detail: "No holdings API. Collateral positions and cash balances can be read."
25	p8 "built on the platform and stayed"	The tenders and proxies show institutions buying and keeping Aladdin modules, not building on it. Only Janus Henderson's SDK changes show building.	"…a buyer wants to know that other institutions chose Aladdin and kept it."
26	p8 and Exhibit 15 row 3, pension fund as the first Studio case study	No record shows a pension fund using Studio. Public bodies are often barred from endorsing vendors (for US federal staff, 5 CFR 2635.702).	Lead with "a client whose engineers have contributed to the SDK". Keep pension funds as an Aladdin reference only, "if they use Studio".

27	Exhibit 8 and p7	The caveat that JPMorgan is also a custodian sits in small print. The disclosure exists because BlackRock owns more than 5% of JPMorgan. A separate information-services line disappears in the 2025 proxy, just as the Aladdin figure jumps.	Add to the note: "Disclosed because BlackRock owns more than 5% of JPMorgan. The proxies do not say what the payments cover; JPMorgan is also a custodian that connects to Aladdin."
28	p5 "move through review states before they are approved"	The states exist, but `reviewState` is writable in the create and update requests, and the spec shows no enforced sequence.	"Compliance rule assignments carry a review state, from draft through review to approved."
29	Exhibit 14 "Approvals on record"	Capital-flow approver IDs can be supplied by the caller (`manualStateCycle`). The stronger control is the user's Aladdin "newcash" permissions and the Open → Authorized → Confirmed states.	"Capital flows need the user's cash permissions and move from open to authorized to confirmed, with approvers recorded."
30	Exhibit 14 client boxes	"Route Aladdin calls through the client's own gateway" rests on one client's changes and is optional. "Run in the client's own environment" leaves out BlackRock-hosted Studio Compute.	"Can route calls through the client's own gateway." "Run in the client's environment or in Studio Compute."
31	Exhibit 11 (p9)	The input guardrail step, which comes first, is missing. The title says "calling production APIs", but the loop also calls agents. "Uses GPT-4" is present tense from May 2025. "Never investment advice" is a product policy, not a pipeline step.	Add "Input guardrails" first. Title: "…then calling tools and agents, most of them production Aladdin APIs". Note: "as of the May 2025 talk". Last box: "One answer; policy: no investment advice".
32	p9 heading "Aladdin's AI works only for logged-in users"	True for Copilot, but it reads as if no outside AI can reach Aladdin.	"Aladdin's own AI assistant works only inside Aladdin. Data providers let outside AI tools reach their data."
33	p9 "So I infer that Copilot and Studio use the same APIs"	There are about 2,000 tools against 54 public APIs. The SDK's core specs are labelled "Internal". None of the 54 answers the talk's exposure example. Some public APIs do back BlackRock's screens (Risk Radar, ADC Admin Center).	"Copilot and Studio's public APIs run on the same Aladdin Graph runtime, and some public APIs back BlackRock's own screens, such as Risk Radar. Whether Copilot's tools use the public APIs is not public; with about 2,000 tools against 54 public APIs, most probably do not."
34	Exhibit 12 rows	SimCorp: a portfolio management agent is live today. Arcesium's MCP can act, not only read. Clearwater's "800 AI agents for clients" were mostly built by clients and its own teams. FactSet's claim is "industry's first production-grade", and LSEG was live seven weeks earlier. Bloomberg's MCP is a Data License product filed under system-of-record rivals.	Reword each row to the source. Move Bloomberg to the data group, or note that its MCP is a data product.
35	Exhibit 17 scoring and legend	State Street's client-only sandbox scores 1, while Aladdin's scores 0. Aladdin's public sample code and notebooks score 1, while Bloomberg's SDK scores 2. Bloomberg's public docs score 1. The AI-connector column scores whether a connector exists, against a "no client relationship" premise. The legend's 0 merges "none" with "closed to non-clients". The test date is "October 8", but some sources are dated October 16 and 20.	Rescore: State Street trial 0, Aladdin sample code 2, Bloomberg docs 2. Add to the note: "0 means not reachable without a client relationship, not that the capability is absent. The AI connector column scores whether a connector exists; most are for licensed clients." Fix the date.
36	p14 "may compete with Aladdin for certain uses"	Snowflake runs Data Cloud, and Databricks is a named connector partner.	Add: "Snowflake and Databricks are included as benchmarks for developer access; Aladdin Data Cloud runs on Snowflake."

37	Exhibit 3 secondary	The controls column runs down past hosting, but those controls apply only to API calls. Data Cloud and ADX are shown only as ways in, not out. The Copilot box should say "tools and agents".	End the controls column at the shared-services row. Show Data Cloud and ADX as two-way. Copilot box: "about two thousand tools and agents".
38	Exhibit 4 Admin Center; Developer Portal; Governance	Admin Center: 4 of 10 operations are "intended for internal BlackRock use only". Portal: only "guides" are tied to it in the sources. Governance: "Data, tools, and governance support" describes all of Studio.	Admin Center: "The Data Cloud dataset screen, partly run by BlackRock." Portal: "Catalog, guides, field dictionary, notebooks and samples; the portal sits behind a client login." Governance: quote only "access management and usage monitoring".
39	Exhibit 6, Goel cell	"Every transaction via an event" and "15 minutes of latency" are aims ("we really want to").	Prefix the cell with "Aims for:".
40	Exhibit 1 "Of eight Aladdin products"	It reads as BlackRock's full lineup. BlackRock lists 4 platforms and 9 solutions. Preqin has no row, though the paragraph names it.	Note: "A selection. BlackRock lists four platforms and nine solutions."
41	Exhibit 16 "after qualification"	Most items in that column come with being a client. The "Get help" row mixes a client's code changes with documentation.	Column: "After becoming a client". Get help, public cell: "The SDK repo's SUPPORT.md has headings but no channels; the maintainers merged a client engineer's changes."
42	Exhibit 10 window (p8)	The Q3 2026 call (October 14) came before the October 19 date. Studio count 0, so the finding holds.	"27 earnings calls, Q1 2020 to Q3 2026".

Tier 4. Precision and nits

#	Where	Problem	Fix
43	p5 long-running	10 of the 35 marked operations are the status calls themselves.	"25 operations run as long-running jobs, and each of the 10 specifications that has them offers a status call."
44	Exhibit 4 / Exhibit 2 caption, events	The SDK ships Studio notification specs (REST, Studio project events).	"No public specification covers business events."
45	p8 "four usage measures"; "fastest"	Two of the four are BlackRock-only counts, and ranking unlike measures means little.	"four volume estimates to 2030".
46	Exhibit 10 "add-on"	The calls say "alongside".	"cited among products clients use alongside Aladdin Enterprise".
47	Exhibit 9 PPF	"Address" is a web-address field. Both figures cover ten years and both lots.	"…up to £74.0 million over ten years, against an estimate of up to £55 million". "Supplier web address".
48	Exhibit 9 PBGC	The two contracts have exactly the same yearly price. The records never name Aladdin.	"…at the same yearly price. The records name BlackRock, not Aladdin."
49	Register 58 title; register 3 date	Source 58's title does not match the notice ("Order management, portfolio monitoring and risk analytics system"). Source 3 is dated only "2025"; it is December 16, 2025.	Correct both entries.
50	p3 revenue	The 24% includes about $210M from Preqin. Organic growth is about 10.5%, and ACV grew 16% excluding Preqin.	Optional: add "about 10 percent without Preqin".

51	p12 "Aladdin Graph … over 200 institutions"	That line is generic BlackRock technology preamble in the posting.	"BlackRock's technology job postings say it builds for BlackRock and over 200 institutions."
52	p12 "Copilot's guardrails are tested daily"	What is tested daily is system-prompt behaviour and end-to-end routing.	"Copilot's system prompts are tested daily by AI judges."
53	Exhibit 7	"Only Woodhead names them all" is near-circular, since the list is his definition. Goel covers developer tooling in substance.	"Only Woodhead lists all three parts in one definition."
54	Exhibit 6 title "on the record"	Studio staff also describe it in writing.	"in recordings"
55	Exhibit 13	Self-selected respondents and no sample size. Poll 3 was a forced choice ("choosing only from these fours").	Note: "Respondents to a BlackRock webinar; number not given." Title: "…would most like to stop maintaining, of four options".
56	Exhibit 15 row 1	Publishing the stability labels shows that 24 of 54 specs are alpha. No public Databricks connector for Aladdin was found.	Keep it if deliberate. "Snowflake (and Databricks, if offered) connectors".
57	Exhibit 16 "See the catalog: public"	The specs are only inside wheel files.	"Public, but only by unpacking the packages".
58	Exhibit 4 note, Studio Compute	The SDK uses both "Aladdin Studio Compute" and "Aladdin Compute".	"The SDK uses both names for the same hosted environment."
59	Exhibit 12 smaller rows	MSCI is index data first. S&P works "via any MCP-enabled AI application". Snowflake's server is broader than Cortex Agents. Databricks is partly in beta. State Street has discussed an "Alpha Co-pilot" in beta.	Add each qualifier in a few words.
60	Exhibit 5 note	5 of the 301 operations are marked internal-only in their own descriptions.	Optional: "five are marked for internal use".
61	Exhibit 4, Spaces	"Studio Space" is in BlackRock's own notification options.	Optional: "Spaces: named alongside Projects; not described publicly."

5. Evidence the report does not use, which supports it

#	Evidence	Where it helps
62	Source 71: Garson, "100%" and "we are hearing that from our clients", on clients wanting Aladdin plug-ins in their own copilots.	The demand step of the evolved thesis; replaces the job posting.
63	Copilot's registry access is set "by environment, user group and app"; calls are made "on behalf of the user".	The shared tool layer and permissions in the evolved thesis.
64	Some public specs back BlackRock's own screens: Risk Radar (RiskRule, RiskConfig), ADC Admin Center.	The shared runtime claim (p9, Exhibit 14).
65	Aladdin user permissions: PermissionAPI (function, action, portfolio); per-operation prerequisites such as permType='brkrSettlm'; capital flows need "newcash" permissions.	The controls story. This is the strongest public control, and the report omits it.
66	The order lifecycle in the spec: PM authorises, then the trader activates; Hold and Do-not-trade statuses; authorisation is read-only via the API.	The "a person approves each action" point (p12) and the rebalance demo (Exhibit 15 row 4).

67	The SDK's `client_credentials` flow and UserAPI "bot accounts".	Technical access for an agent (fix 8).
68	10-K: "BlackRock is empowering clients with data and APIs"; Investor Day slides 33 and 35.	The investor-facing platform story, replacing the 17x attribution.
69	SimCorp's Launchpad gives client-built agents "controlled access … within a common governance framework", with an audit trail.	An honest competitive comparison for the evolved thesis: a rival is moving the same way.